r_tutorial_week_05

Werkcollege 5

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How to use this tutorial

Welcome back! In Week 4 you learned to **test** a claim with a t-test. You compared one mean to a known value, and you compared the means of two groups.

This week we look at **relations between two variables**. For example: *do older students study fewer hours?* The tool we use is **simple regression**. You will see that regression is not a completely new tool: the two-group t-test from Week 4 is actually a special case of it.

We work in two parts:

1. **A number by two groups**: we compare a t-test with a regression that uses a **dummy variable**.

2. **A number by a number**: we draw a scatterplot, fit a regression line, and check the assumptions.

Work through this page carefully, from top to bottom. Every short explanation is followed by a **Your turn** task to do yourself.

General tips:

- Read each part slowly and in order. Each step builds on the one before it.
- Type the code yourself instead of copying and pasting it. You learn faster this way.
- Errors are normal. Read the message and try again.
- If you are stuck, check the FAQ or ask a teacher.
- Save your work often.

---

1 Start workflow

We begin, as always, with the **start workflow**:

1. Open **RStudio**: click on the RStudio application.
2. Set the **working directory**: click *Session* > *Set Working Directory* > *Choose Directory*, and choose your `research_methods` folder.
3. Open a **new script**: click *File* > *New File* > *R Script*.
4. On the first line, write a comment with the title, for example: `# R tutorial - Week 5`.
5. Save the script as `week5.R` in your `research_methods` folder.
6. Load your cleaned data from Week 2 into an object called `survey`:

```
survey <- read.csv("class_survey_clean.csv")
```

> ⚠ Watch out
>
> If you see `Error: cannot open file ...`, R cannot find the file. Check that your working directory is `research_methods` (run `getwd()` to print it), and that `class_survey_clean.csv` really sits in that folder.

Your turn

Complete the **start workflow** and load `survey`. Check that `survey` appears in the **Environment** pane (top-right).

---

2 The five steps, once more

In Week 4 you learned that every hypothesis test follows the **same five steps**. This week we use them again, now for regression. Keep this list next to you: you will use it in every test for the rest of the course.

> **The five steps**
>
> 1. **Hypotheses** – write down the null and the alternative.
> 2. **Assumptions** – check that the test is allowed for your data.
> 3. **Test statistic** – compute the t-value.
> 4. **p-value** – find how surprising that t-value is if the null hypothesis were true.
> 5. **Interpretation** – decide about the null hypothesis and say what it means **in context**.

Step 5 deserves a reminder. Statistics does **not** give you an automatic, "objective" answer. A p-value and an estimate are **tools**. You still have to **interpret** them yourself: How big is the effect, in real units? Does it matter for practice? How good is the sample? Two researchers can look at the same output and write slightly different, but both reasonable, conclusions, as long as they explain their reasoning.

---

3 Part 1 – A number by two groups: t-test and regression

We start with a question you already know from Week 4: *do Pedagogy and Psychology students differ in self-efficacy?* We answer it twice: once with a t-test, and once with regression. Then we compare the two outputs.

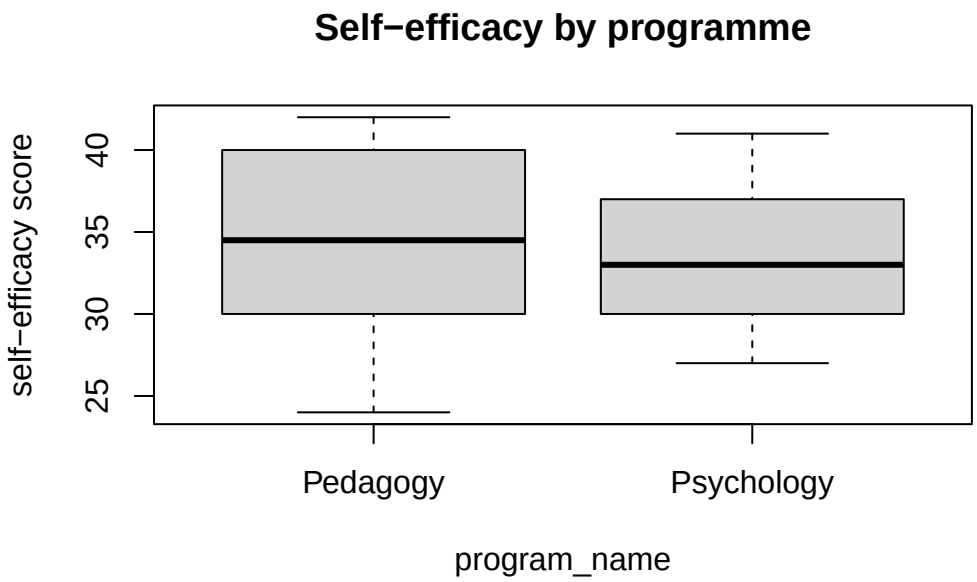
We keep only the two programmes, exactly like last week:

```
two_programs <- survey[survey$program_name %in% c("Pedagogy", "Psychology"), ]
```

3.1 Look first: a boxplot and the group means

Before any test, always **look** at the data:

```
boxplot(self_efficacy ~ program_name,
        data = two_programs,
        main = "Self-efficacy by programme",
        ylab = "self-efficacy score")
```



Now the group means and the **raw difference** between them. "Raw" means: in the original units of the scale (here, scale points), not converted into some other unit.

```
mean_ped <- mean(two_programs$self_efficacy[two_programs$program_name == "Pedagogy"])
mean_psy <- mean(two_programs$self_efficacy[two_programs$program_name == "Psychology"])

mean_ped
#> [1] 33.9
mean_psy
#> [1] 33.55556
mean_psy - mean_ped   # the raw difference: Psychology minus Pedagogy
#> [1] -0.3444444
```

The raw difference is the easiest effect size to understand: "Psychology students score, on average, this many points higher (or lower) than Pedagogy students."

Your turn

Make the boxplot and compute the two group means and the raw difference for **your own** two groups. Write down the raw difference. Is it large or small compared to the range of the scale?

3.2 The t-test (from Week 4)

We run the two-sample t-test. This time we add `var.equal = TRUE`. This tells R to assume that both groups have the **same spread** in the population. We do this here **only** because regression makes the same assumption, so the two outputs will match exactly. (In Week 4 we used Welch's test, which does not need this assumption.)

```
t.test(self_efficacy ~ program_name, data = two_programs, var.equal = TRUE)
#>
#>  Two Sample t-test
#>
#> data:  self_efficacy by program_name
#> t = 0.13488, df = 17, p-value = 0.8943
#> alternative hypothesis: true difference in means between group Pedagogy and group Psychol
#> 95 percent confidence interval:
#>  -5.043285  5.732174
#> sample estimates:
#>   mean in group Pedagogy mean in group Psychology
#>                 33.90000                 33.55556
```

Write down three numbers from this output: the **t-value**, the **p-value**, and the two **group means**.

3.3 Making a dummy variable

Regression works with numbers. So we turn the group into a number with a **dummy variable**. A dummy variable has only two values:

- `0` for the **reference group** (here: Pedagogy),
- `1` for the other group (here: Psychology).

We make it with `ifelse()` from Week 2:

```
two_programs$psychology <- ifelse(two_programs$program_name == "Psychology", 1, 0)
table(two_programs$program_name, two_programs$psychology)   # check: every Pedagogy row is 0
#>
#>              0  1
#>   Pedagogy   10  0
#>   Psychology  0  9
```

> **Example: reading a dummy**
>
> A dummy is named after the group that gets the `1`. So `psychology = 1` means "this person studies Psychology", and `psychology = 0` means "this person does **not** study Psychology" (here: studies Pedagogy).

> **Going further (optional): R can make the dummy for you**
>
> If you put a text variable straight into a regression, R makes the dummy itself. It picks the reference group alphabetically (here: Pedagogy). So `lm(self_efficacy ~ program_name, data = two_programs)` gives the same result. Making the dummy by hand helps you see what happens.

3.4 The regression with the dummy

We fit a regression model with the function `lm()`, which stands for **linear model**. The formula form `y ~ x` is the same as in `boxplot()` and `t.test()`: the **dependent variable** (y) comes before the `~`, the **independent variable** (x) after it.

```
model_dummy <- lm(self_efficacy ~ psychology, data = two_programs)
summary(model_dummy)
#>
#> Call:
#> lm(formula = self_efficacy ~ psychology, data = two_programs)
#>
#> Residuals:
#>    Min     1Q Median     3Q    Max
#> -9.900 -3.900  0.100  4.272  8.100
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  33.9000     1.7575  19.288 5.41e-13 ***
#> psychology   -0.3444     2.5536  -0.135    0.894
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 5.558 on 17 degrees of freedom
#> Multiple R-squared:  0.001069,   Adjusted R-squared:  -0.05769
#> F-statistic: 0.01819 on 1 and 17 DF,  p-value: 0.8943
```

Look at the table under `Coefficients:`. It has two rows:

- `(Intercept)`: the predicted score when the dummy is `0`. That is the **mean of Pedagogy**.
- `psychology`: how much the prediction changes when the dummy goes from `0` to `1`. That is the **raw difference** between the two means (Psychology minus Pedagogy).

Check this yourself:

```
coef(model_dummy)     # intercept and slope
#> (Intercept)  psychology
#>  33.9000000  -0.3444444
c(mean_ped, mean_psy - mean_ped)   # Pedagogy mean, and the raw difference
#> [1] 33.9000000 -0.3444444
```

The numbers are the same.

3.5 Comparing the two outputs

Now compare the t-test and the regression side by side:

What	t-test output	regression output
Mean of Pedagogy	`mean in group Pedagogy`	`(Intercept)` estimate
Raw difference	difference of the two means	`psychology` estimate
t-value	`t`	`t value` in the `psychology` row
p-value	`p-value`	`Pr(>|t|)` in the `psychology` row

The t-value has the **same size** in both outputs. The **sign** can be different, because `t.test()` subtracts in the order Pedagogy minus Psychology, and our dummy measures Psychology minus Pedagogy. The p-value is exactly the same.

So a two-group t-test (with equal spreads) and a regression with a dummy are **the same test**. Regression is the more general tool: next you will see that it also works when x is a number instead of a group.

Your turn

For **your own** two groups:

1. Run `t.test()` with `var.equal = TRUE`.
2. Make a dummy variable with `ifelse()` and check it with `table()`.
3. Run `lm()` with the dummy and look at `summary()`.
4. Copy the table above into your script as comments, and fill in your own numbers in both columns.

5. Write a conclusion of two or three sentences. Use the **raw difference** (in scale points), the p-value, and say what the difference means for practice.

---

4 Part 2 – A number by a number: simple regression

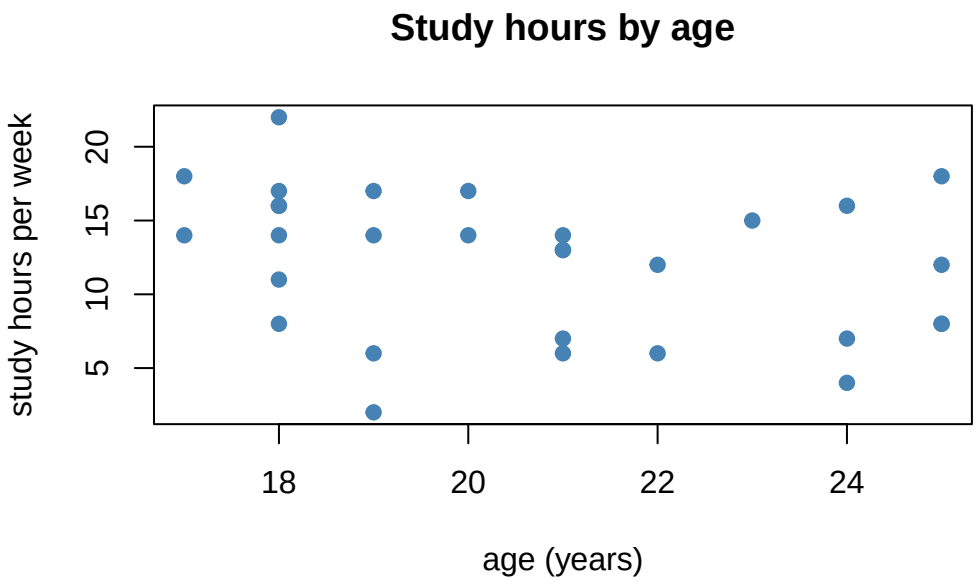
Now x is a number. Our example question: *do older students study fewer hours per week?*

- **Dependent variable** (y): `study_hours`, the variable we want to explain.
- **Independent variable** (x): `age`, the variable we use to explain it.

4.1 Look first: a scatterplot

For two numeric variables, the right picture is a **scatterplot**: one dot per person, with x on the horizontal axis and y on the vertical axis. We use `plot()` with the same `y ~ x` formula:

```
plot(study_hours ~ age,
     data = survey,
     pch = 19, col = "steelblue",
     main = "Study hours by age",
     xlab = "age (years)",
     ylab = "study hours per week")
```



When you look at a scatterplot, ask three questions:

1. **Direction**: do the dots go up (positive), down (negative), or neither?
2. **Form**: do they follow a straight line, or a curve?
3. **Strength and odd points**: are the dots close to a line, or widely spread? Are there dots far away from the rest?

Your turn

Choose **two numeric variables** for your own research question: one dependent variable (y) and one independent variable (x). Make a scatterplot with a title and clear axis labels. Answer the three questions above in one sentence each.

4.2 A short recap: correlation

You met the **correlation** (r) in the lecture. It is a number between -1 and $+1$ that describes how strongly two numeric variables move together **in a straight line**:

- r close to $+1$: a strong positive relation (higher x, higher y).
- r close to -1: a strong negative relation (higher x, lower y).
- r close to 0: no straight-line relation.

```r
cor(survey$age, survey$study_hours)
#> [1] -0.3014262
```

A correlation can also be tested, with $H_0 : \rho = 0$ (no relation in the population; ρ is the Greek letter "rho", the correlation in the population). The function `cor.test()` does this with a t-test:

```r
cor.test(survey$age, survey$study_hours)
#>
#>  Pearson's product-moment correlation
#>
#> data:  survey$age and survey$study_hours
#> t = -1.6728, df = 28, p-value = 0.1055
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  -0.59687775  0.06601149
#> sample estimates:
#>       cor
#> -0.3014262
```

Remember the `t`, `df`, and `p-value` here. You will see the same numbers again in a moment.

Your turn

Compute the correlation for your two variables with `cor()`, and test it with `cor.test()`. Is the correlation positive or negative? Is it strong or weak?

4.3 From correlation to a regression line

The correlation tells you **how strongly** two variables are related. Regression tells you **how much** y changes when x goes up by one unit. It does this by drawing the best straight line through the dots:

$$\hat{y} = a + b \cdot x$$

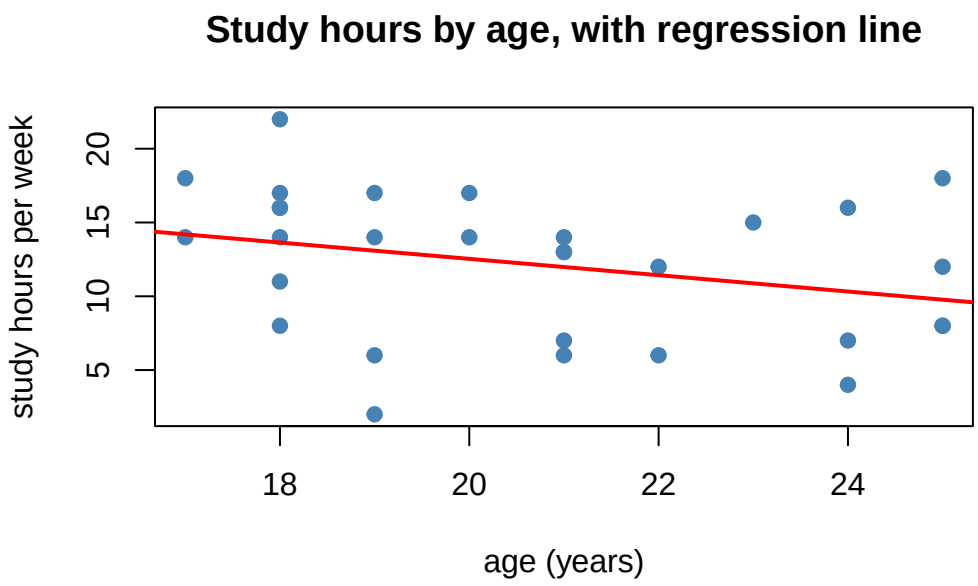
- $\hat{y}$ ("y-hat") is the **predicted** value of y.
- a is the **intercept**: the predicted y when x is 0.
- b is the **slope** (or **regression coefficient**): how much the predicted y changes when x goes up by **1**.

We fit the model with `lm()`, exactly like in Part 1:

```
model <- lm(study_hours ~ age, data = survey)
coef(model)
#> (Intercept)         age
#>  23.6000000  -0.5532258
```

We draw the line on top of the scatterplot with `abline()`:

```
plot(study_hours ~ age,
     data = survey,
     pch = 19, col = "steelblue",
     main = "Study hours by age, with regression line",
     xlab = "age (years)",
     ylab = "study hours per week")
abline(model, col = "red", lwd = 2)
```



Read the slope in words, always in the units of your variables. Here the slope is -0.55: "for every year older, the predicted study time changes by -0.55 hours per week." Because the slope is negative, older students are predicted to study **less**. A positive slope would mean they are predicted to study **more**. This raw slope is the effect size for regression: it tells you how big the relation is, in real units.

> ⚠ Watch out: the intercept is not always meaningful
>
> The intercept is the predicted y when x is **0**. Here that means the study hours of a student aged 0, which makes no sense. The intercept is needed to draw the line, but you often do not interpret it. In Part 1 it **was** meaningful, because x = 0 meant "Pedagogy".

Your turn

Fit `lm()` for your own two variables and draw the regression line on your scatterplot. Write the slope in one sentence, in the real units of your variables (for example "for every extra hour of …, the score goes up by … points").

4.4 Predicted values and residuals

For every person, the model gives a **predicted value** $\hat{y}$: the point on the line at that person's x. The person's real score is usually not exactly on the line. The distance between the real score and the predicted score is the **residual**:

$$\text{residual} = y - \hat{y}$$

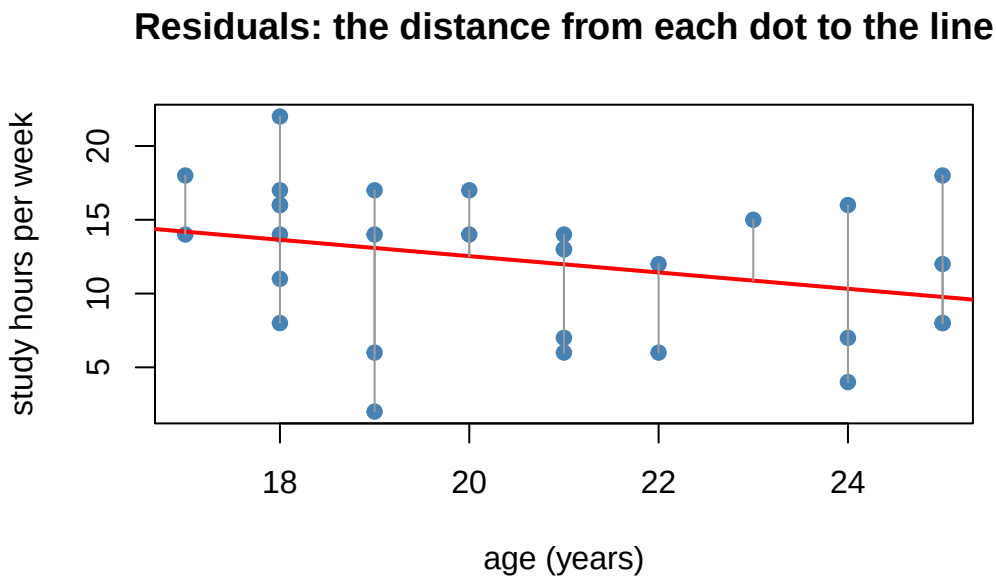
- A **positive** residual: the person is **above** the line (scores higher than predicted).
- A **negative** residual: the person is **below** the line (scores lower than predicted).

R gives you both with `fitted()` and `residuals()`:

```
head(data.frame(age       = survey$age,
                observed  = survey$study_hours,
                predicted = round(fitted(model), 1),
                residual  = round(residuals(model), 1)))
#>   age observed predicted residual
#> 1  25        8       9.8     -1.8
#> 2  17       18      14.2      3.8
#> 3  22        6      11.4     -5.4
#> 4  20       14      12.5      1.5
#> 5  21       13      12.0      1.0
#> 6  18       16      13.6      2.4
```

In a picture, the residuals are the vertical lines between each dot and the regression line:

```r
plot(study_hours ~ age,
     data = survey,
     pch = 19, col = "steelblue",
     main = "Residuals: the distance from each dot to the line",
     xlab = "age (years)",
     ylab = "study hours per week")
abline(model, col = "red", lwd = 2)
segments(x0 = survey$age, y0 = fitted(model),
         x1 = survey$age, y1 = survey$study_hours,
         col = "grey60")
```



The regression line is the line that makes these residuals as **small as possible** (to be exact: it makes the sum of the squared residuals as small as possible). This is why the method is called **least squares**.

Your turn

Look at the first row of the table above. Check by hand (in R, as a calculator) that the residual is the observed value minus the predicted value. Then use `coef(model)` to compute the predicted value yourself: `a + b * age`.

---

5 Testing the slope: the five steps

Is the relation in our sample strong enough to say something about the population? We walk the five steps.

5.1 Step 1 – Hypotheses

We test whether the slope in the **population** (β, the Greek letter "beta") is zero:

- $H_0 : \beta = 0$ – there is **no** linear relation between age and study hours in the population.
- $H_1 : \beta \neq 0$ – there **is** a linear relation (positive or negative).

This is a **two-sided** test, our default, just like in Week 4.

> **i** A special case (optional): one-sided tests
>
> If you have a strong reason to expect one direction **before** you look at the data (for example "older students study **less**", $H_1 : \beta < 0$), you could test one-sided. `summary()` always reports the two-sided p-value. For a one-sided test you halve it, but **only** if the slope goes in the direction you predicted. In this course, use two-sided unless you have a very good reason.

Your turn

Write down H_0 and H_1 for your own two variables, in words **and** with β.

5.2 Step 2 – Assumptions

Regression has more assumptions than the t-test. We discuss them in a separate section below, because checking them takes some work. For now, remember the list:

1. **Linearity**: the relation between x and y is a straight line.
2. **Independent observations**: one row per person, and people do not influence each other.
3. **Normality of the residuals**: the residuals are roughly normally distributed.
4. **Homoscedasticity**: the spread of the residuals is about the same for all values of x.
5. **No measurement error in x**: the independent variable is measured (almost) perfectly.

We come back to all five in the section *Checking the assumptions*.

5.3 Step 3 – The test statistic

The t-value for the slope works like every t-value you have seen: the estimate, divided by its **standard error**.

$$t = \frac{b - 0}{SE_b}$$

The "-0" is there because H_0 says the slope is 0. Both b and its standard error are in the `summary()` output:

```
summary(model)
#>
#> Call:
#> lm(formula = study_hours ~ age, data = survey)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -11.0887  -3.1524   0.9645   3.1081   8.3581
#>
#> Coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  23.6000     6.8897   3.425  0.00191 **
#> age          -0.5532     0.3307  -1.673  0.10551
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 4.754 on 28 degrees of freedom
#> Multiple R-squared:  0.09086,    Adjusted R-squared:  0.05839
#> F-statistic: 2.798 on 1 and 28 DF,  p-value: 0.1055
```

In the `Coefficients:` table, the `age` row gives `Estimate` (b), `Std. Error` (SE_b), `t value`, and `Pr(>|t|)` (the p-value). We compute the t-value by hand from the table:

```
coef_table <- summary(model)$coefficients
coef_table        # the coefficients table on its own
#>               Estimate Std. Error   t value    Pr(>|t|)
#> (Intercept) 23.6000000  6.8897452  3.425381 0.001912969
#> age         -0.5532258  0.3307184 -1.672800 0.105505297

b    <- coef_table["age", "Estimate"]
se_b <- coef_table["age", "Std. Error"]
```

```r
t_value <- (b - 0) / se_b
t_value
#> [1] -1.6728
```

> **Where does the standard error come from?**
>
> The standard error of the slope is the **width of the sampling distribution of the slope**, just like the standard error of the mean in Week 3. If you drew many samples, each would give a slightly different slope. SE_b tells you how much the slope wobbles from sample to sample. R computes it for you; you do not need the formula.

Your turn

Run `summary()` for your own model. Find b and SE_b in the table, and compute the t-value by hand. Check that it matches the `t value` in the output.

5.4 Step 4 – The p-value

The t-value follows a t-distribution with $n - 2$ degrees of freedom (we lose 2: one for the intercept, one for the slope). We use `pt()` exactly like in Week 4:

```r
n  <- nrow(survey)
df <- n - 2
p_value <- 2 * pt(-abs(t_value), df = df)   # two-sided p-value
p_value
#> [1] 0.1055053
```

> **Correlation and regression give the same test**
>
> Compare the `t`, `df`, and `p-value` with the output of `cor.test()` above. They are **identical**. Testing "the correlation is 0" and testing "the slope is 0" is the same test.

Your turn

Compute the two-sided p-value for your slope with `pt()`. Check that it matches `Pr(>|t|)` in `summary()` and the p-value from `cor.test()`. On paper, you would look it up in Table B with $df = n - 2$.

5.5 A confidence interval for the slope

As in Week 3, a single estimate is not the whole story. A 95% **confidence interval** for the slope gives a range of plausible values for β in the population:

```
confint(model)
#>                 2.5 %   97.5 %
#> (Intercept)  9.486997 37.71300
#> age         -1.230672  0.12422
```

Read the `age` row. If the interval contains 0, then "no relation" is still a plausible value, and the two-sided test is not significant at 0.05. If 0 is outside the interval, the test is significant. Testing and confidence intervals are two sides of one coin, just as in Week 4.

Your turn

Compute `confint()` for your model. Does the interval for the slope contain 0? Does this agree with your p-value?

5.6 How much does the model explain? R-squared

The slope tells you **how much** y changes per unit of x. Another question is: **how well** does x predict y? Do the dots lie close to the line, or far away? This is measured by **R-squared** (R^2).

R^2 is the **share of the differences in y** (the variance) that the model explains. It lies between 0 and 1:

- $R^2 = 0$: x explains nothing of y.
- $R^2 = 1$: x explains y perfectly (all dots on the line).

```
summary(model)$r.squared
#> [1] 0.09085777
```

Multiply by 100 to read it as a percentage: "age explains about 9% of the differences in study hours between students." The other part is explained by other things that are not in the model.

In simple regression, R^2 is simply the correlation squared:

```r
cor(survey$age, survey$study_hours)^2
#> [1] 0.09085777
```

> **i** Raw slope or R-squared?
>
> Both are useful, but they answer different questions. The **raw slope** says how big the change in y is, in real units: easy to judge for practice. R^2 (like the correlation) is **standardized**: it has no units, which makes it easier to compare across studies, but harder to judge for practice. In this course we focus on the raw slope and use R^2 as extra information.

Your turn

Compute R^2 for your model, and check that it equals your correlation squared. Write one sentence: "x explains about …% of the differences in y."

5.7 Step 5 – Interpretation

Now we put everything together. A good conclusion answers these questions:

1. **The test:** is p smaller than 0.05? Do we reject H_0 or not?
2. **The size:** what is the raw slope, in real units? What is the confidence interval?
3. **The meaning:** is this relation large enough to matter in practice? What does it mean for the people you study?
4. **The limits:** how good is the sample? Are the assumptions met? Are there influential observations (see below)?

> **i** Example: a conclusion
>
> "In our sample of 30 students, older students studied somewhat fewer hours: for every year older, about 0.6 hour less per week (95% CI from -1.2 to 0.1). This relation was **not** statistically significant ($t(28)$ = -1.67, p = 0.106), so we do not reject H_0. The interval is wide and includes both no relation and a relation of more than one hour per year. With only 30 students we cannot say much about the population. A larger sample is needed before we draw any conclusion for practice."

Notice what this example does **not** say: it does not say "there is no relation". Not rejecting H_0 is not the same as proving H_0. And a significant result would not automatically mean an important result: with a very large sample, even a slope of a few minutes per year could become significant. You always have to judge the **size** of the effect in the **context** of your research question.

Your turn

Write a conclusion of four to six sentences for your own model. Answer the four questions above. Use the raw slope (in real units), the confidence interval, the t-value, and the p-value.

---

6 Checking the assumptions

A regression always gives you numbers, even when the model is wrong for your data. So we must check the assumptions **before** we trust the t-value and the p-value. Most checks are done with **pictures of the residuals**.

6.1 Two assumptions you check by thinking

Some assumptions cannot be seen in a plot. You check them by thinking about **how the data were collected**.

- **Independent observations.** Each person should appear only once, and people should not influence each other's answers. This is violated, for example, when you measure the same person twice, or when classmates fill in the questionnaire together.
- **No measurement error in x.** Regression assumes that the independent variable is measured (almost) perfectly. `age` is measured very precisely. A questionnaire score, such as self-efficacy, always contains some measurement error. Measurement error in x makes the slope look **smaller** than it really is.

Your turn

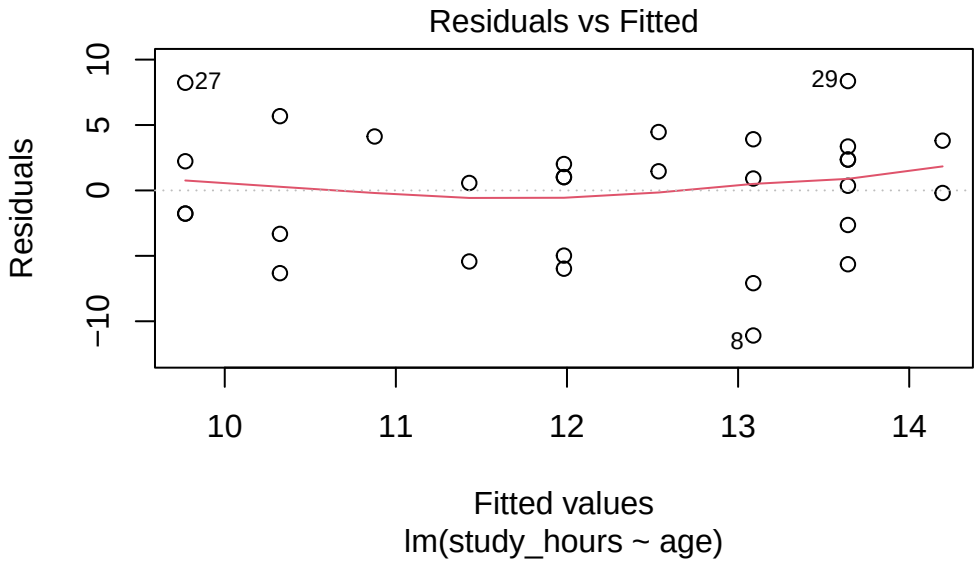
For your own model, write one or two sentences on each of these two assumptions. How were your data collected? How precisely is your x measured?

6.2 Three assumptions you check with plots

R makes the diagnostic plots for you with `plot(model)`. We look at two of them.

Plot 1 – Residuals vs Fitted: linearity and homoscedasticity. This plot shows the predicted values on the x-axis and the residuals on the y-axis:

```r
plot(model, which = 1)
```

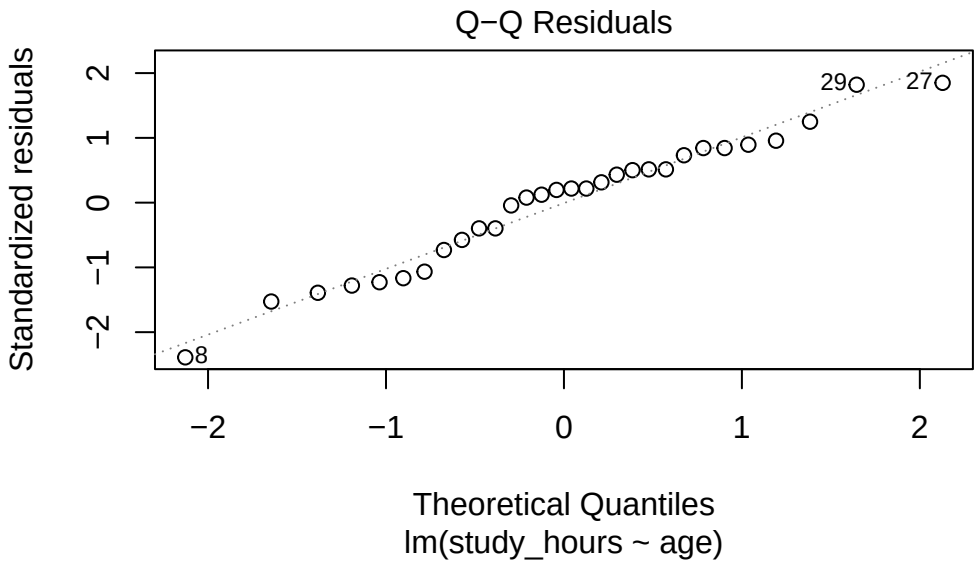


What you want to see:

- **Linearity:** the dots are spread around the horizontal line at 0 **without a pattern**. The red line should be roughly flat. A clear curve (a U-shape or a hill) means the relation is not a straight line.
- **Homoscedasticity:** the vertical spread of the dots is about the **same** from left to right. A funnel shape (narrow on one side, wide on the other) means the spread is not equal: this is called **heteroscedasticity**.

Plot 2 – Q-Q plot: normality of the residuals. This plot compares the residuals with a perfect normal distribution:

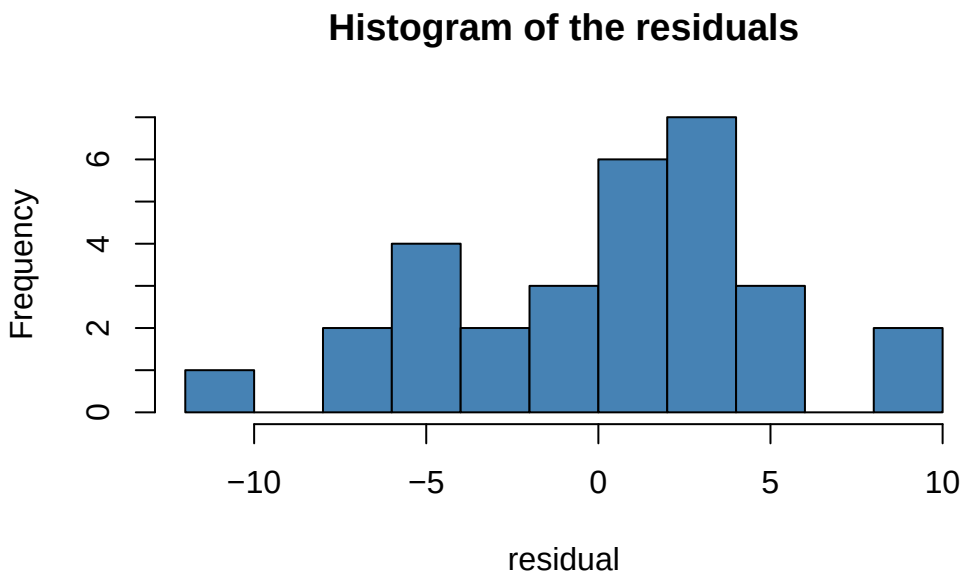
```r
plot(model, which = 2)
```



What you want to see: the dots lie roughly **on the dashed line**. Small wiggles at the ends are normal. Dots that curve far away from the line, especially at both ends, point to non-normal residuals.

You can also simply make a histogram of the residuals, as in Week 4:

```
hist(residuals(model),
     breaks = 10,
     col = "steelblue",
     main = "Histogram of the residuals",
     xlab = "residual")
```



Watch out: normality of the residuals, not of x or y

The normality assumption is about the **residuals**, which is equivalent to the normality assumption about the distribution of y given x. The normality assumption is **not** about x or y themselves. A skewed x or y variable is not a problem in itself. Always check the residuals.

Small samples are hard to judge

With only 30 people, the plots always look a bit messy. Do not look for perfection. Look for **clear** patterns: a strong curve, a clear funnel, or dots far away from the rest. When in doubt, describe what you see and be careful in your conclusion.

Your turn

Make the two diagnostic plots and the histogram of the residuals for your own model. For each of the three assumptions (linearity, homoscedasticity, normality of the residuals), write one sentence: what do you see, and is the assumption reasonable?

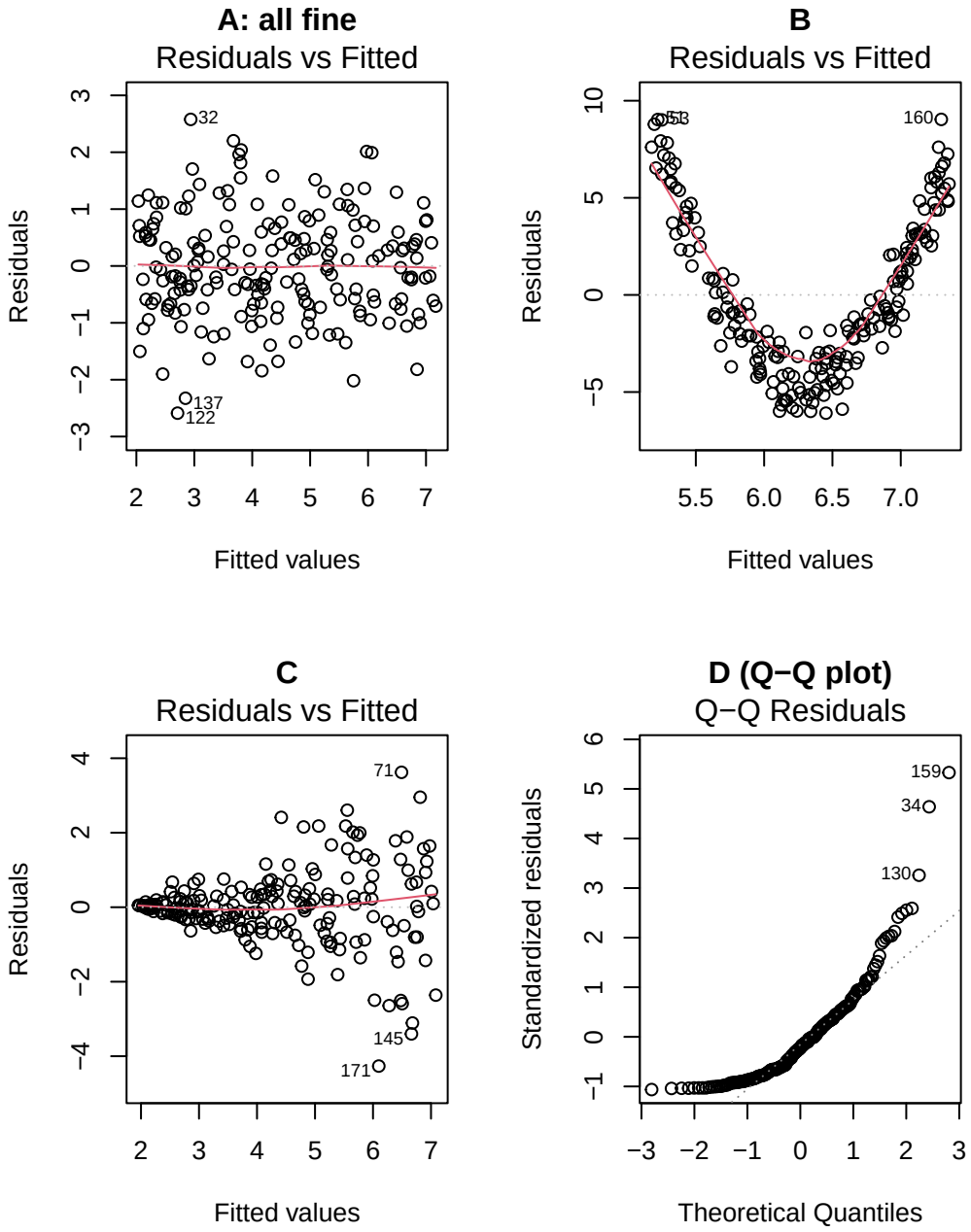
6.3 What do violations look like?

With real data of 30 people, it is hard to learn what a "bad" plot looks like. So we **simulate** four datasets, as in Week 3: one where all assumptions hold, and three where one assumption is broken. This is code to run and read; you do **not** need to write it yourself.

```
set.seed(2026)
n_sim <- 200
x <- runif(n_sim, 0, 10)                                  # x values between 0 and 10

y_good   <- 2 + 0.5 * x + rnorm(n_sim, sd = 1)            # all assumptions hold
y_curve  <- 2 + 0.5 * (x - 5)^2 + rnorm(n_sim, sd = 1)    # the relation is a curve
y_funnel <- 2 + 0.5 * x + rnorm(n_sim, sd = 0.2 * x)      # the spread grows with x
y_skew   <- 2 + 0.5 * x + (rexp(n_sim, rate = 1) - 1)     # skewed residuals

par(mfrow = c(2, 2))                                      # four plots in one window
plot(lm(y_good   ~ x), which = 1, main = "A: all fine")
plot(lm(y_curve  ~ x), which = 1, main = "B")
plot(lm(y_funnel ~ x), which = 1, main = "C")
plot(lm(y_skew   ~ x), which = 2, main = "D (Q-Q plot)")
```




```
par(mfrow = c(1, 1))                                    # back to one plot per window
```

Your turn

Plot A shows what "all fine" looks like. For plots B, C, and D, decide which assumption is violated: **linearity**, **homoscedasticity**, or **normality of the residuals**. Explain in one sentence per plot what you see that tells you this.

Hint: compare each plot with plot A. Where is the pattern, a curve, a funnel, or dots leaving the line?

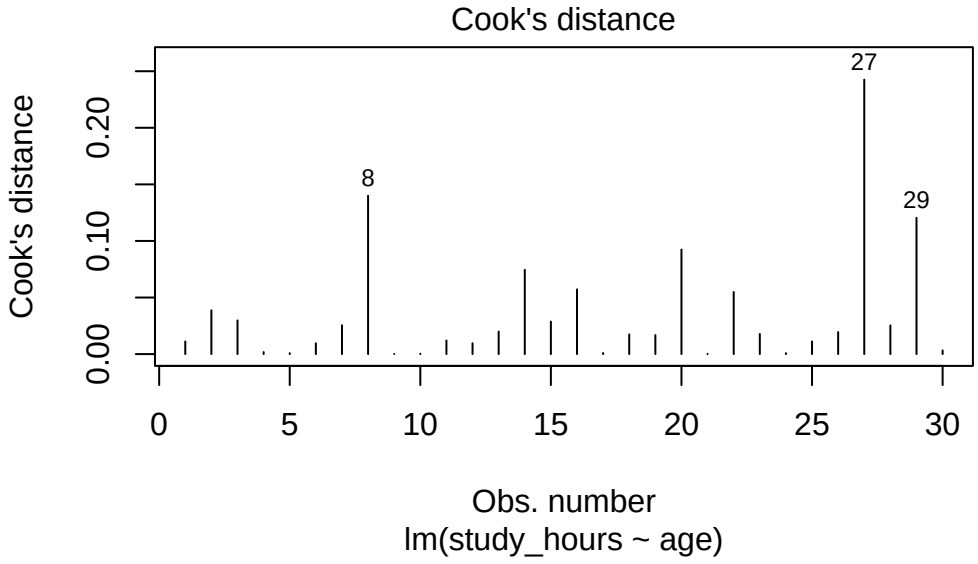
---

7 Influential observations

Sometimes **one person** has a large effect on the regression line. If you removed that one person, the line (and maybe your conclusion) would change a lot. Such a person is called an **influential observation**.

A person is usually influential when they have an **unusual x value** (far from the other x values) **and** do not follow the pattern of the others. We measure influence with **Cook's distance**: one number per person. The larger, the more influence.

```
plot(model, which = 4)   # Cook's distance for each person
```



A common rule of thumb: look more closely at people with a Cook's distance larger than $4/n$.

```
cooks <- cooks.distance(model)
cut_off <- 4 / nrow(survey)
cut_off
#> [1] 0.1333333

which(cooks > cut_off)            # the row numbers above the rule of thumb
#>  8 27
#>  8 27
survey[cooks > cut_off, c("age", "study_hours")]   # their values
#>    age study_hours
#> 8   19           2
#> 27  25          18
```

7.1 Re-running the analysis without them

To see how much these people matter, we run the model **again without them** and compare. We do **not** delete them from `survey`; we make a new object:

```
survey_check <- survey[cooks <= cut_off, ]     # everyone at or below the rule of thumb
model_check  <- lm(study_hours ~ age, data = survey_check)

# compare the two slopes and p-values
rbind(all_people     = summary(model)$coefficients["age", ],
      without_influ  = summary(model_check)$coefficients["age", ])
#>                 Estimate Std. Error   t value    Pr(>|t|)
#> all_people    -0.5532258  0.3307184 -1.672800 0.105505297
#> without_influ -0.8446677  0.2977908 -2.836447 0.008720827
```

Compare the two rows. Did the slope change much? Did the p-value cross 0.05? Did the conclusion change?

In our example, two students change the result: without them, the slope is steeper and the p-value drops below 0.05. So **two people** decide whether the result is "significant" or not. This is exactly the situation where you must be careful, as the box below explains.

> **!** Never remove people just to get a better result
>
> An influential observation is **not** automatically a mistake. Only remove a person when there is a **clear reason**, for example a typing error (an age of 190) or someone who does

not belong to your population. If there is no such reason, keep everyone in the main analysis and **report both results**: "Without these two students the slope was …, so our conclusion depends (or does not depend) on these two people." Removing people until $p < 0.05$ is a serious research error.

Your turn

For your own model:

1. Make the Cook's distance plot and find the people above $4/n$.
2. Look at their values. Is there a clear reason to remove them (for example, an impossible value)?
3. Re-run the model without them and compare the slope and the p-value.
4. Write two sentences: does your conclusion change, and what would you report?

---

8 Recap: what you can now do

Last week you tested means. This week you learned to describe and test **relations between two variables**. You can now:

- apply the **five steps** of a hypothesis test to regression;
- make a **dummy variable** with `ifelse()` and show that a regression with a dummy gives the same test as a t-test;
- read the **intercept** and **slope** of a dummy regression as a group mean and a **raw mean difference**;
- make a **scatterplot** and compute and test a **correlation** with `cor()` and `cor.test()`;
- fit a **simple regression** with `lm()`, draw the line with `abline()`, and interpret the **slope** in real units;
- explain **predicted values** and **residuals**;
- compute the **t-value** of the slope by hand, find the **p-value**, and compute a **confidence interval** with `confint()`;
- compute and interpret **R-squared**;
- name the five **assumptions** of regression and check them with `plot(model)`;
- find **influential observations** with Cook's distance, and re-run the analysis to see whether your conclusion depends on them;
- write a conclusion that combines the test, the size of the effect, and the **context**.

A small cheat sheet to keep:

You want to…	You write…
Make a dummy variable	`df$d <- ifelse(df$group == "B", 1, 0)`
t-test with equal spreads	`t.test(y ~ group, data = df, var.equal = TRUE)`
Scatterplot	`plot(y ~ x, data = df)`
Correlation	`cor(df$x, df$y)`
Test a correlation	`cor.test(df$x, df$y)`
Fit a regression	`model <- lm(y ~ x, data = df)`
Full output	`summary(model)`
Intercept and slope	`coef(model)`
Draw the line	`abline(model)`
Predicted values / residuals	`fitted(model)`, `residuals(model)`
t-value of the slope by hand	`b / se_b`
Two-sided p-value	`2 * pt(-abs(t), df = n - 2)`
Confidence interval for the slope	`confint(model)`
R-squared	`summary(model)$r.squared`
Linearity and homoscedasticity	`plot(model, which = 1)`
Normality of the residuals	`plot(model, which = 2)`
Cook's distance	`plot(model, which = 4)`, `cooks.distance(model)`

Well done. Keep your materials in the `research_methods` folder, we build directly on them next week.
